Mariano Marcos State University
SDG 15 · Life on Land

STORYBOARD · CHAPTER 08 OF 13

08

Identifying and monitoring threatened species

3 frames

THE STORY

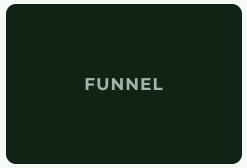
Knowing what is at risk: two decades of work identifying, conserving and monitoring threatened indigenous food plants.

MMSU'S COMMITMENT

Take an active role in protecting and promoting sustainable use of terrestrial ecosystems, responsible land use, and biodiversity conservation.

THE FILMSTRIP · 3 FRAMES

2025 Ongoing programme



1


2


3

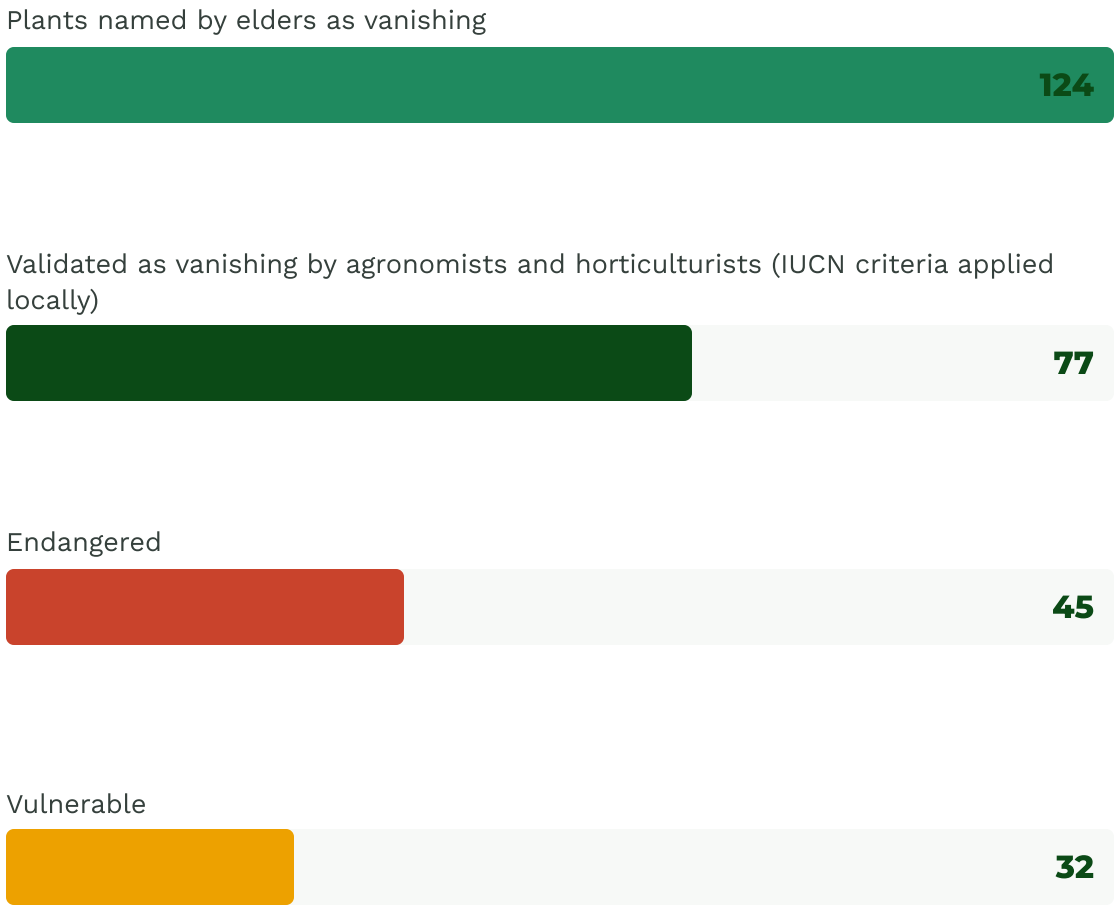
AT A GLANCE

77
plants validated vanishing

45 + 32
endangered + vulnerable

1st
RDC Region 1 R&D item

2003 survey of Laoag City, Batac, Pasuquin, San Nicolas and Dingras



2003 AND 2011 · IDENTIFYING THE PLANTS AT RISK PART 1 OF 2

From Vanishing Plants: How MMSU Identified Threatened Indigenous Food Plants

1. **2003 · THE SURVEY**
   MMSU researchers (Dr. Miriam Pascua, Ms. Menisa Antonio, Dr. Dionisio Bucao) documented plants elders saw as vanishing; 77 of 124 were validated, among them sugod-sugod (Momordica cochinchinensis), kapas-kapas and bignay (Antidesma bunius).
2. **2011 · THE STUDY**
   SEARCA published “Survey and Characterization of Indigenous Food Plants in Ilocos Norte”: 46 plants from 27 families in seven upland and remote municipalities, with identity, uses and locations.
3. **METHOD NOTE**
   The 2003 classification applied IUCN criteria locally; it is not the IUCN Red List.

77	46	27
validated vanishing	IFPs in 7 towns	plant families

Source: The Philippine Star, “MMSU researchers identify vanishing plant species,” 14 September 2003 · Antonio, M. A., Utrera, R. T., Agustin, E. O., Jamias, D. L., Badar, A. J. and Pascua, M. E. (2011), “Survey and Characterization of Indigenous Food Plants in Ilocos Norte, Philippines,” SEARCA Agriculture and Development Discussion Paper No. 2011-2, 61 pp. · MMSU Research Directorate, “MMSU research on indigenous food plants wins best paper at SUC-ACAP 2025 Conference,” 29 August 2025 *+3 more on the last page*



Farmers and local officials trained on alukon (himbabao) propagation, Brgy. 11 Sta. Rosa, Sarrat, 2 May 2025. (Photo: MMSU)



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

2025 · CONSERVING AND MONITORING PART 2 OF 2

To Regional Policy: The IFP Program Conserves and Monitors the Plants

1. **THE PROGRAM**
   The IFP Program, led by Dr. Menisa A. Antonio of the Research Directorate, documents IFP species and protects them through a botanical garden (ex situ), a himbabao (alukon) genebank, circa situ cultivation of wild yams, and community foodscapes; it is advancing omics-based molecular characterization.
2. **IN 2025**
   Trained 64 farmers and officials in Sarrat on alukon propagation (2 May), extended seedbanks and foodscapes to new Vintar partner barangays (Oct), and won Best Paper at the SUC-ACAP 2025 Biennial Conference.
3. **REGIONAL POLICY**
   The first R&D initiative adopted by the Regional Development Council of Region 1, providing input for policy advocacy on conserving and using these species; propagation continues (PNA, 6 Jun 2026).

64
farmers trained

1st
RDC Region 1 R&D initiative

Source: The Philippine Star, “MMSU researchers identify vanishing plant species,” 14 September 2003 · Antonio, M. A., Utrera, R. T., Agustin, E. O., Jamias, D. L., Badar, A. J. and Pascua, M. E. (2011), “Survey and Characterization of Indigenous Food Plants in Ilocos Norte, Philippines,” SEARCA Agriculture and Development Discussion Paper No. 2011-2, 61 pp. · MMSU Research Directorate, “MMSU research on indigenous food plants wins best paper at SUC-ACAP 2025 Conference,” 29 August 2025 *+3 more on the last page*



Capture: Philippine News Agency post of 6 June 2026 on indigenous food plants propagated at the MMSU NRCMC botanical garden (photos: Leilanie Adriano, PNA).



Partners receive indigenous food plant planting materials for the Community Foodscapes and Seedbanks, World Food Day 2025.

6 JUN 2026 · MMSU NATURAL RESOURCES CONSERVATION AND MANAGEMENT CENTER, BATAC

Indigenous Food Plants Propagated at the MMSU Botanical Garden Are Shared with Schools and Communities

Ongoing programme · Philippine News Agency, 6 June 2026

1. **THE REPORT**
   The Philippine News Agency reported that sugod-sugod, alokon, buga and karot are being propagated at the NRCMC botanical garden.
2. **THE SHARING**
   Planting materials go to schools and community stakeholders for food gardens, to boost food security and give farmers an additional income source.
3. **THE LINK**
   The garden is the ex situ arm of the IFP Program, which in 2025 extended community foodscapes and seedbanks to new partner barangays.

Source: Philippine News Agency, Facebook post on indigenous food plants at the MMSU Natural Resources Conservation and Management Center botanical garden, 6 June 2026

WHERE THE STORY LANDS

From a 2003 survey to a regional programme, MMSU has identified vanishing plants, kept them in a botanical garden, a genebank and community seedbanks, and brought the work into regional policy.

77	45 + 32	1st
plants validated vanishing	endangered + vulnerable	RDC Region 1 R&D item

THE FRAMES

1. **From Vanishing Plants: How MMSU Identified Threatened Indigenous Food Plants**
   2003 and 2011
2. **To Regional Policy: The IFP Program Conserves and Monitors the Plants**
   2025
3. **Indigenous Food Plants Propagated at the MMSU Botanical Garden Are Shared with Schools and Communities**
   6 Jun 2026

SOURCES (6)

1. The Philippine Star, “MMSU researchers identify vanishing plant species,” 14 September 2003
   https://philstar.com/business/2003/09/14/220344/mmsu-researchers-identify-vanishing-plant-species
2. Antonio, M. A., Utrera, R. T., Agustin, E. O., Jamias, D. L., Badar, A. J. and Pascua, M. E. (2011), “Survey and Characterization of Indigenous Food Plants in Ilocos Norte, Philippines,” SEARCA Agriculture and Development Discussion Paper No. 2011-2, 61 pp.
   https://www.searca.org/pubs/discussion-papers?pid=130
3. MMSU Research Directorate, “MMSU research on indigenous food plants wins best paper at SUC-ACAP 2025 Conference,” 29 August 2025
   https://research.mmsu.edu.ph/news/details/mmsu-research-on-indigenous-food-plants-wins-best-paper-at-suc-acap-2025-conference
4. MMSU News, “MMSU trains Sarrat farmers on alukon propagation,” 9 May 2025
   https://www.mmsu.edu.ph/news/mmsu-trains-sarrat-farmers-on-alukon-propagation
5. MMSU Extension Directorate, “MMSU expands indigenous food plant seedbanks and foodscapes in LGUs,” October 2025
   https://extension.mmsu.edu.ph/news/show/389/mmsu-expands-indigenous-food-plant-seedbanks-and-foodscapes-in-lgus
6. Philippine News Agency, Facebook post on indigenous food plants at the MMSU Natural Resources Conservation and Management Center botanical garden, 6 June 2026
   https://www.facebook.com/pnagovph/posts/1411199747723142/